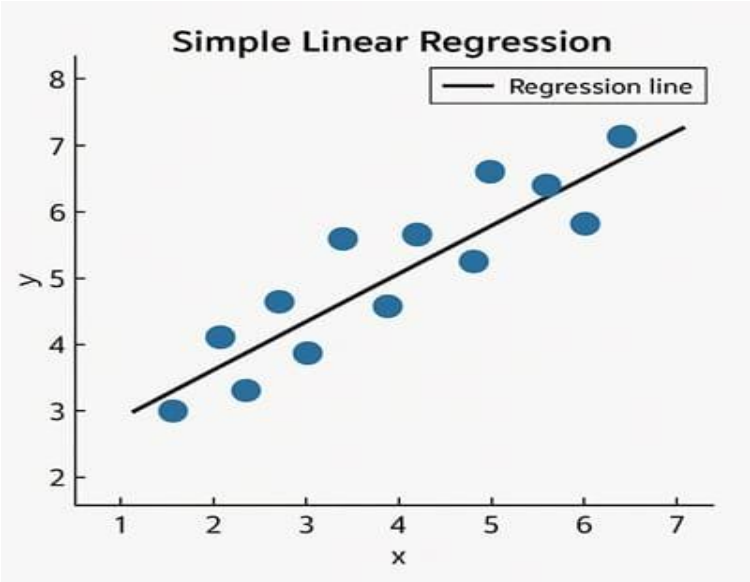
Overview of Classification Algorithms in Machine Learning

1. Linear Regression

Description:

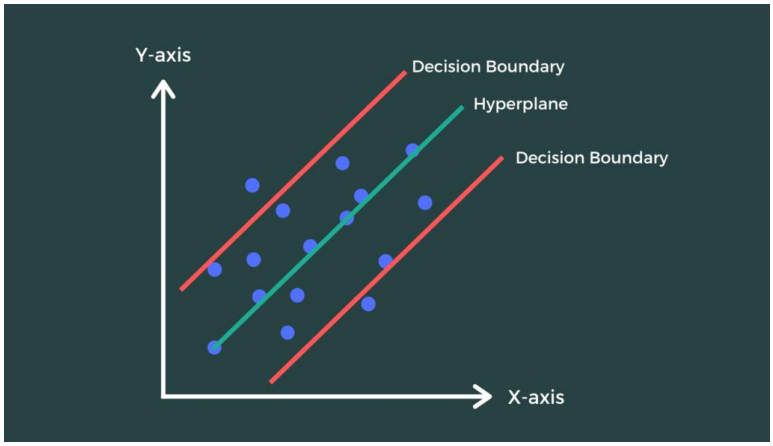
Linear Regression is a simple supervised learning algorithm used for predicting continuous values. It finds a straight-line relationship between input features and output. In house price prediction, it estimates price based on features like bedrooms, sqft, etc. It is fast and easy but works best when data has a linear pattern.



2. Support Vector Regression (SVR)

Description:

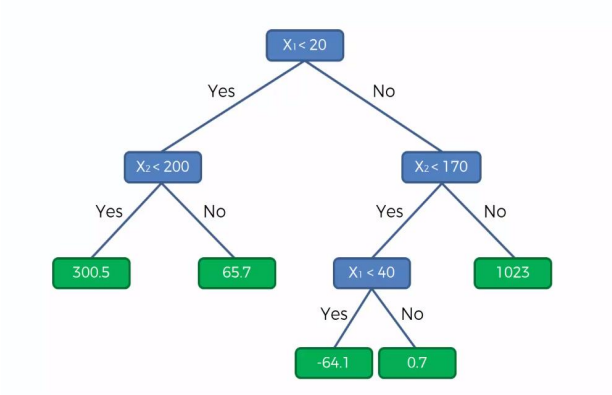
SVR is the regression version of Support Vector Machine. It tries to fit the best line within a margin (called epsilon tube) while minimizing error. It is powerful for complex data but sensitive to scaling. In this project, it helps model non-linear relationships between house features and price.



3. Decision Tree Regressor

Description:

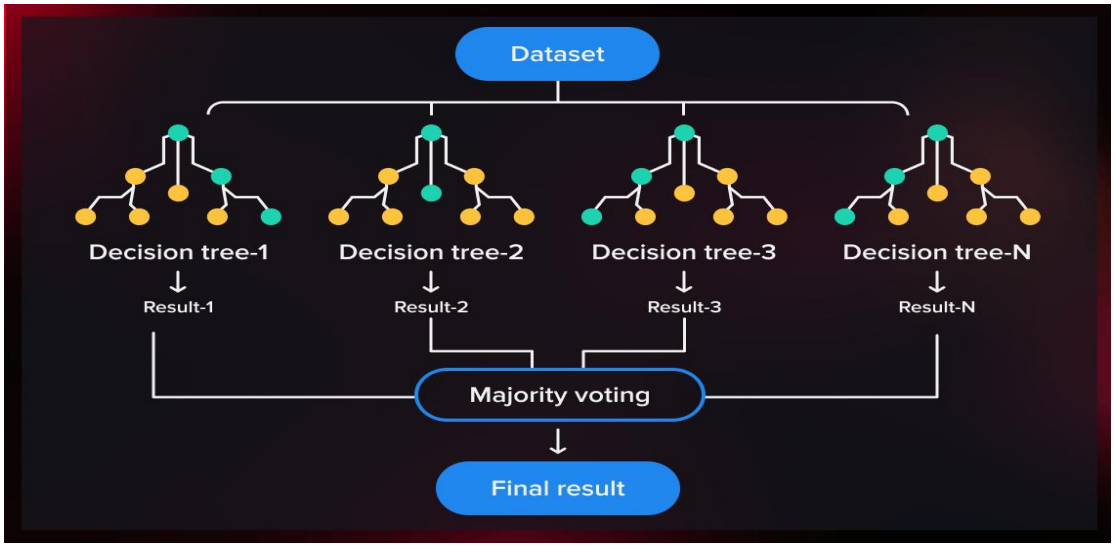
A Decision Tree Regressor splits data into branches based on feature values. Each split reduces error and leads to a final prediction at the leaf node. It is easy to understand but may overfit if the tree becomes too deep.



4. Random Forest Regressor

Description:

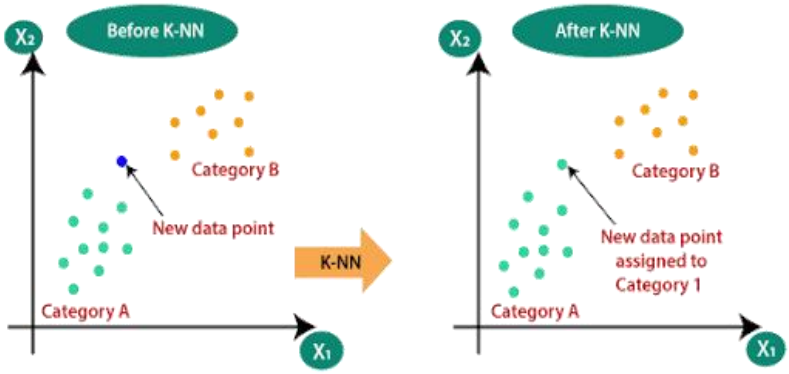
Random Forest is an ensemble method that combines multiple decision trees. Each tree gives a prediction, and the final output is the average of all. It reduces overfitting and improves accuracy, making it one of the best models for this dataset.



5. K-Nearest Neighbors (KNN) Regressor

Description:

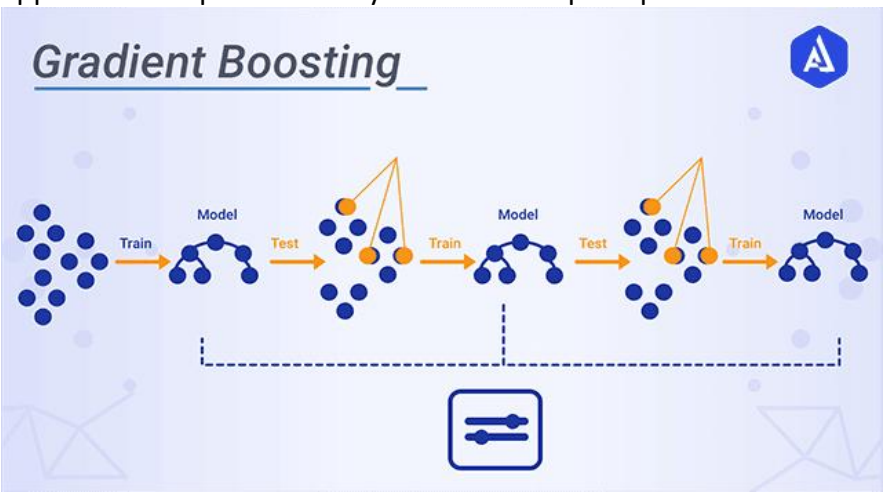
KNN predicts house price based on the average of nearest similar houses. It uses distance (like Euclidean distance) to find similar data points. It is simple but slower for large datasets.



6. Gradient Boosting Regressor

Description:

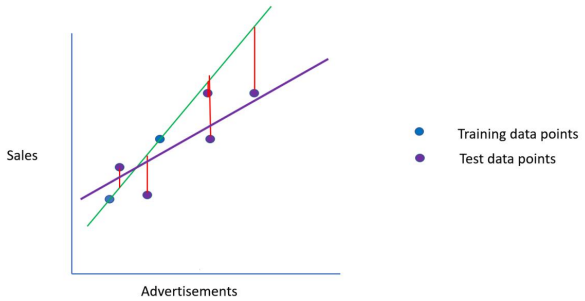
Gradient Boosting builds models sequentially, where each new model corrects errors of the previous one. It gives very high accuracy and is widely used in real-world applications. It performs very well in house price prediction.



7. Ridge Regression

Description:

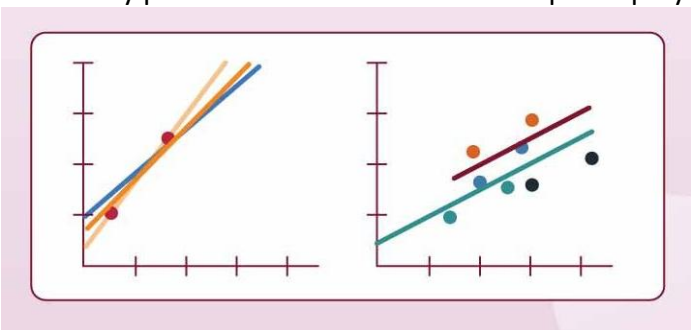
Ridge Regression is an improved version of Linear Regression. It adds a penalty (regularization) to reduce overfitting. It works well when features are correlated and improves model stability.



8. Lasso Regression

Description:

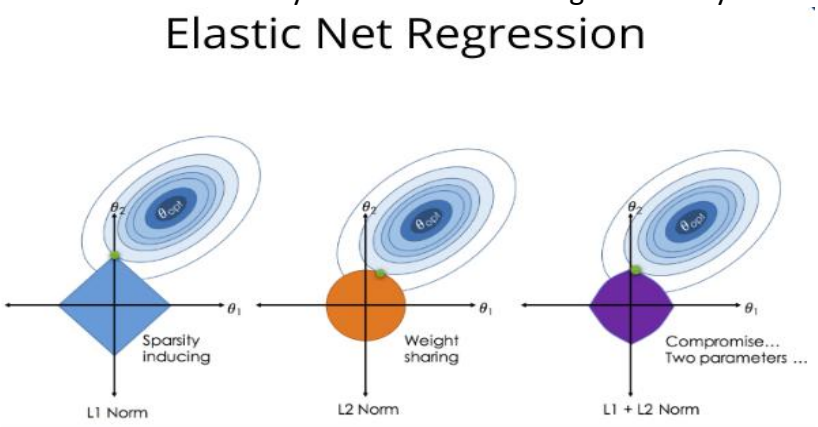
A linear model with L1 regularization that can shrink some coefficients to zero. Effectively performs feature selection. Helps simplify the model.



9. ElasticNet Regression

Description:

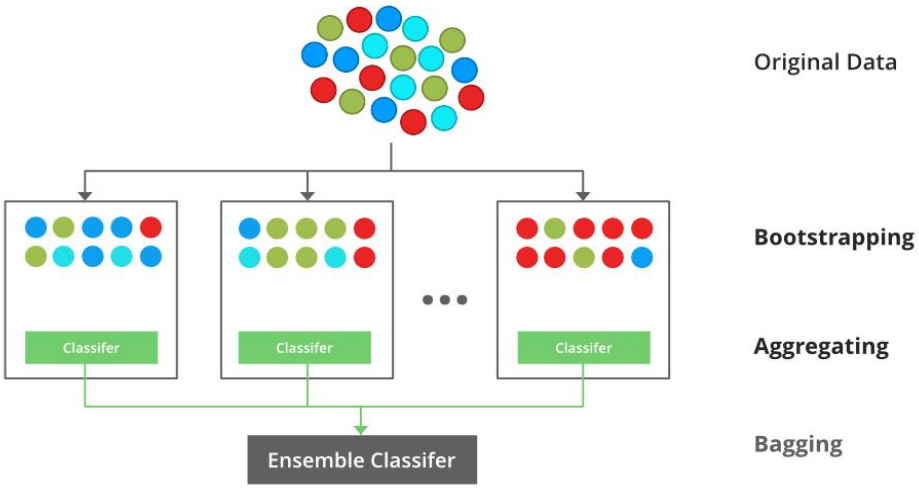
Combines both L1 (Lasso) and L2 (Ridge) regularization. Balances feature selection and coefficient stability. Useful when dealing with many correlated features.



10. XGBoost Regression

Description:

An advanced boosting algorithm that builds trees sequentially with optimization techniques. Handles missing values and complex patterns efficiently. Widely used in competitions.



Final Conclusion:

In this project, multiple regression algorithms were applied to predict house prices using features such as bedrooms, bathrooms, and square footage. The models were evaluated using the R^2 score, and their performance varied based on their learning approaches. Ensemble methods like Random Forest and Gradient Boosting achieved better accuracy by combining multiple models and reducing errors. Linear models were simpler but less effective for complex patterns. Overall, proper model selection and parameter tuning are essential for achieving the best prediction performance.

What I have Learned:

- I learned how machine learning can be used for real-world prediction problems like house prices.
- I understood different regression algorithms and how they work.
- I learned the importance of feature scaling and train-test splitting.
- I understood how models are compared using performance metrics.
- I learned that ensemble methods give better performance than simple models.
- Overall, I gained practical knowledge of building and evaluating a complete machine learning pipeline.